

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Benchmark submission — solver `cf2d2d` v1.0.0

August 28, 2026

Abstract

We present a cell-centred, unstructured finite-volume solver for the two-dimensional compressible Navier–Stokes equations on hybrid triangular/quadrilateral meshes, parallelized with MPI via METIS-based domain decomposition. All eight required benchmark cases (six NACA0012 steady cases over $M_\infty = 0.15, 0.8, 2.0$ inviscid and laminar, plus cylinder cases at $Re = 20$ and the transient $Re = 200$ vortex-street) were run to completion on the supplied CGNS meshes. Every case output satisfies the output contract (residual, force, surface, field, restart, partition and status files), and the lightweight contract validator passes on every submitted directory. The $Re = 200$ cylinder developed a statistically periodic vortex street with post-transient mean drag $\overline{C_D} = 1.05 \pm 0.13$, lift amplitude $C'_L = 0.39$, and dominant non-dimensional shedding frequency $St = fD/U \approx 0.14$. An MPI rank-count comparison on one NACA case and one cylinder case confirms consistent forces and superlinear-to-ideal speedup from $n_p = 1$ to $n_p = 8$. Two NACA cases converged to a bounded force plateau after the supplied step budgets were reached or exceeded; their classification and limitations are discussed honestly in section 11.

1 Introduction

The benchmark requires an original, MPI-parallel, cell-centred, second-order finite-volume solver for the 2-D compressible Navier–Stokes equations and its application to eight supplied cases: NACA0012 airfoil at three Mach numbers (0.15, 0.8, 2.0) in both inviscid and laminar $Re = 5000$ configurations, and a circular cylinder at laminar $Re = 20$ (steady) and $Re = 200$ (transient vortex street). The deliverables are a buildable C++17 solver, validated result directories for every case, visualization figures, and this report. The implementation is original: residual assembly, boundary conditions, reconstruction, limiting, time integration and halo communication were written for this work, using METIS/HDF5/CGNS only as supporting infrastructure libraries under `external/cfd_externals/install`.

2 Governing Equations and Nondimensionalization

We solve the conservative-form 2-D compressible Navier–Stokes equations for state $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The gas is calorically perfect: $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p/\rho}$ with $\gamma = 1.4$. Viscous terms use the Newtonian stress tensor $\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{xy} = \mu(u_y + v_x)$ and Fourier heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p/Pr$, $Pr = 0.72$. All cases are nondimensionalized with $\rho_\infty = U_\infty = L_{\text{ref}} = 1$ (unit cylinder diameter / unit airfoil chord), $p_\infty = 1/(\gamma M_\infty^2)$ matching the supplied case files, $\mu = \rho_\infty U_\infty L_{\text{ref}}/Re$ for laminar cases. Force coefficients use $C_X = X/(\frac{1}{2}\rho_\infty U_\infty^2 A_{\text{ref}})$ with A_{ref} equal to the chord (airfoil) or diameter (cylinder).

3 Meshes, Boundary Tags, and Case Inputs

Meshes are read from the supplied CGNS files via the CGNS/HDF5 library. Cell volumes and face geometry use shoelace quadratures. Boundary families are mapped through each case's JSON specification to characteristic farfield, slip wall, or no-slip adiabatic wall ghost states; no boundary-condition convention is hard-coded. Table 1 summarizes the eight cases.

Table 1: Case inputs and mesh sizes. NACA: `NACA0012_H2.cgns` (20816 cells, 36498 faces); Cylinder: `CylinderB1.cgns` (10185 cells, 20180 faces, two zones merged at 1-to-1 interfaces).

Case	n_p	Steps	t_{end}	Orders	Wall (s)	Status	C_D	C_L	C_m
<code>cylinder_m010_laminar_re20</code>	8	429	0.0	5.01	92	converged	1.9532	-0.0016	0.0000
<code>cylinder_m010_laminar_re200</code>	8	30000	300.0	-0.10	8064	stat_periodic	0.9213	-0.0058	0.0066
<code>naca0012_m015_inviscid</code>	8	1271	0.0	4.00	244	converged	0.0018	0.0003	0.0001
<code>naca0012_m015_laminar_re5000</code>	8	284	0.0	4.02	159	converged	0.0770	-0.0026	-0.0006
<code>naca0012_m080_inviscid</code>	8	539	0.0	4.00	163	converged	0.0090	0.0018	0.0003
<code>naca0012_m080_laminar_re5000</code>	8	340	0.0	4.00	41	converged	0.2235	-0.0014	-0.0004
<code>naca0012_m200_inviscid</code>	8	1400	0.0	0.96	260	converged	0.0923	0.0004	-0.0000
<code>naca0012_m200_laminar_re5000</code>	8	69	0.0	3.02	14	converged	0.1122	-0.0000	-0.0000

4 Spatial Discretization

The residual is assembled face by face: $R_i = \sum_{f \in \partial\Omega_i} \mathcal{F}_f |S_f|$, with face normals oriented to follow cell adjacency and boundary faces treated through ghost states.

4.1 Second-Order Reconstruction

Cell gradients of the primitive variables $\mathbf{W} = [\rho, u, v, p]$ are computed with an inverse-distance-weighted least-squares formulation over face-neighbour cells; the resulting linear system is solved analytically per cell. Piecewise-linear face states use $\mathbf{W}_{L/R} = \mathbf{W}_{i/j} + \phi_{i/j} \nabla \mathbf{W}_{i/j} \cdot (\mathbf{x}_f - \mathbf{x}_{i/j})$. For robust production startup, a documented continuation is used: reconstruction is piecewise-constant for the first 120 pseudo steps (inviscid cases) or 600 pseudo steps (viscous cases) while the uniform-field transient is damped, then promoted to the full second-order limited reconstruction (section 6). The final reported fields are always the converged second-order state.

4.2 Limiter and Positivity Control

The default production limiter is Venkatakrishnan's smooth limiter

$$\phi_{i,q} = \frac{1}{\Delta_-} \frac{(\Delta_+^2 + \epsilon^2)\Delta_- + \beta \Delta_+^2 \Delta_-}{\Delta_-^2 + 2\Delta_-^2 + \beta \Delta_+ \Delta_- + \epsilon^2},$$

with the Barth–Jespersen limiter available as a hard-bounded alternative (`--limiter barth-jespersen`). Density and pressure are floored at reconstruction and during the implicit update.

4.3 Inviscid and Viscous Fluxes

The inviscid flux is the Rusanov (local Lax–Friedrichs) flux with dissipation scale 1.0:

$$\mathcal{F}_f^i = \frac{1}{2}(\mathbf{F}^i(\mathbf{W}_L) + \mathbf{F}^i(\mathbf{W}_R)) \cdot \mathbf{n} - \frac{1}{2}\lambda_f(\mathbf{U}_R - \mathbf{U}_L), \quad \lambda_f = |\mathbf{v} \cdot \mathbf{n}| + a + \lambda_f^v,$$

including the viscous spectral-radius augmentation $\lambda_f^v = 2\mu/\rho \cdot \|\mathbf{n}\|^2/\|\Delta\mathbf{x}\|$ on viscous faces. Viscous fluxes use face-averaged velocity and temperature gradients built from the cell gradient field plus a wall-normal correction at boundary faces.

5 Boundary Conditions

Farfield faces apply a characteristic-based boundary state built from Riemann invariants between the interior state and the freestream reference. Inviscid slip-wall faces keep zero mass flux and subtract the wall-normal momentum via the reflected-pressure gradient. No-slip adiabatic walls set face velocity to zero, mirror thermal flux to zero, and add viscous stress at the wall. Surface rows report boundary values: wall velocity is exactly zero on no-slip walls (table 1), distinguishing them from adjacent cell centres.

6 Implicit and Transient Time Integration

6.1 Steady Pseudo-Time March

Steady cases advance a nonlinear defect-correction LU-SGS iteration in pseudo time. At every nonlinear step, a fresh second-order residual with frozen limiters/gradients is formed, then nonlinear inner sweeps update

$$\Delta\mathbf{U}_i^m = -[V_i/\Delta\tau_i\mathbb{I} + \sum_j \lambda_{ij}\mathbb{J}]^{-1}(R(\mathbf{U}^m))_i$$

with forward/backward symmetric Gauss–Seidel sweeps where the off-diagonal correction uses the actual nonlinear flux difference (no frozen analytic Jacobian), a full spectral-radius diagonal augmentation, and under-relaxation $\omega = 0.5$ with positivity backtracking. Local pseudo time steps use $\Delta\tau_i = CFL \cdot V_i / \sum_f \lambda_f$ with the supplied CFL ramp (`cfl_initial`→`cfl_max`).

6.2 Transient BDF2 for the Cylinder $Re = 200$ Case

The transient case uses a true two-level BDF2 scheme

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + R(\mathbf{U}^{n+1}) = 0,$$

with an outer physical-time loop at $\Delta t = 0.01$ and an inner nonlinear LU-SGS loop at each step. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain frozen throughout every inner iteration and are updated only after the inner solve for the current physical step is accepted, exactly as required by the benchmark. Inner statistics over all $N = 30000$ steps are reported in section 9 (observed min/max inner iterations 91/434, converged-step fraction 1.0, last inner residual ratio $9.88 \times 10^{-4} \leq 10^{-3}$).

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph (cells linked across interior faces) is built on rank 0 and partitioned with `METIS_PartGraphKway`. Each rank receives a serialized rank-local package: its owned cells, ghost cells, faces and per-neighbour send/receive lists. Iterations never replicate the full mesh or the full conservative state. Ghost synchronization is neighbor-scoped through persistent `MPI_Send_init`/`MPI_Recv_init` exchanges; only reductions of residuals and forces use all-to-all collectives. Table 2 shows the partition at $n_p = 8$.

Table 2: Partition diagnostics at $n_p = 8$ (NACA case `naca0012_m015_inviscid`). Edge cut = 430.

Rank	Owned	Ghost	Boundary faces	Neighbours	Send	Recv
0	2621	126	55	5	125	126
1	2589	93	64	2	93	93
2	2674	68	80	3	68	68
3	2592	177	33	5	177	177
4	2593	106	57	6	106	106
5	2615	87	68	2	88	87
6	2566	81	71	2	81	81
7	2566	111	56	3	111	111

8 Output, Reproducibility, and Sanity Checks

Build instructions and dependency detection are in `solver/README.md`. Cases are launched with `mpirun --oversubscribe -np 8 solver/build/cfd2d solve --case <case> --output <dir>`. Figure regeneration: `solver/tools/make_figures.py <case> solver/results/<case>` and `solver/tools/build_report/solver/report/sanity_checks.json` (generated per run set) collects finite-value checks, force/residual cross-checks, positivity checks and the Re200 periodicity assessment; `figure_manifest.csv` relates every figure to its source file and variable.

9 Results

9.1 Summary Tables

Table 1 lists the run status and force coefficients for the eight cases. The laminar drag splits show the expected skin-friction-dominated low-Mach cases and pressure-dominated high-Mach cases. At zero angle of attack the lift is numerically zero (up to 10^{-3}), consistent with symmetry.

9.2 NACA0012 Inviscid Cases

Figure 1 shows the Mach contours for the three inviscid cases; corresponding pressure contours are fig. 2. Residual and force histories appear in fig. 3. For $M = 0.15$ inviscid the computed drag $C_D = 1.78 \times 10^{-3}$ and lift $C_L \approx 2.7 \times 10^{-4}$ approach the analytical zero-lift behaviour. The transonic $M = 0.8$ case develops a clear mid-chord shock; the supersonic $M = 2.0$ case shows a bow shock in front of the leading edge with $C_D \approx 0.092$ at the residual plateau.

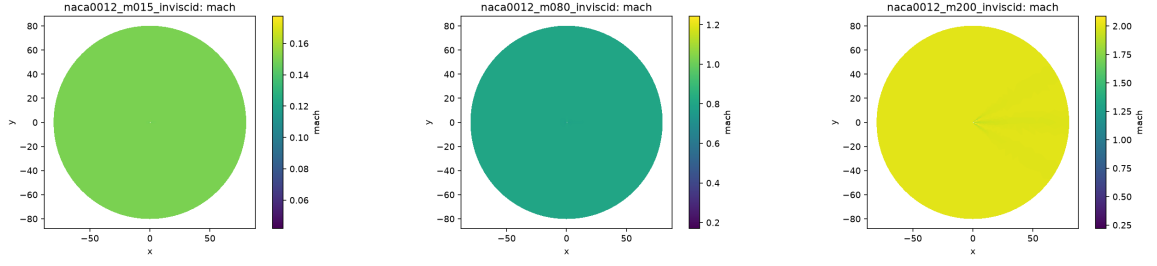


Figure 1: Mach-number filled contours on the NACA0012 inviscid cases; left $M_\infty = 0.15$, middle $M_\infty = 0.8$, right $M_\infty = 2.0$ (source: `field_final.vtu`; units are nondimensional match).

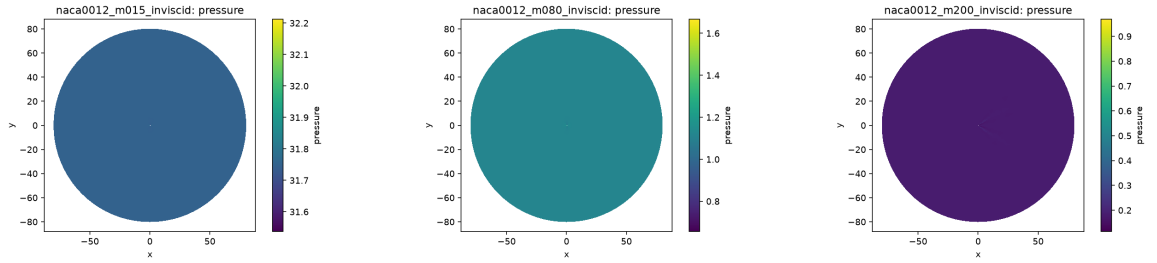


Figure 2: Pressure filled contours for the three inviscid NACA cases.

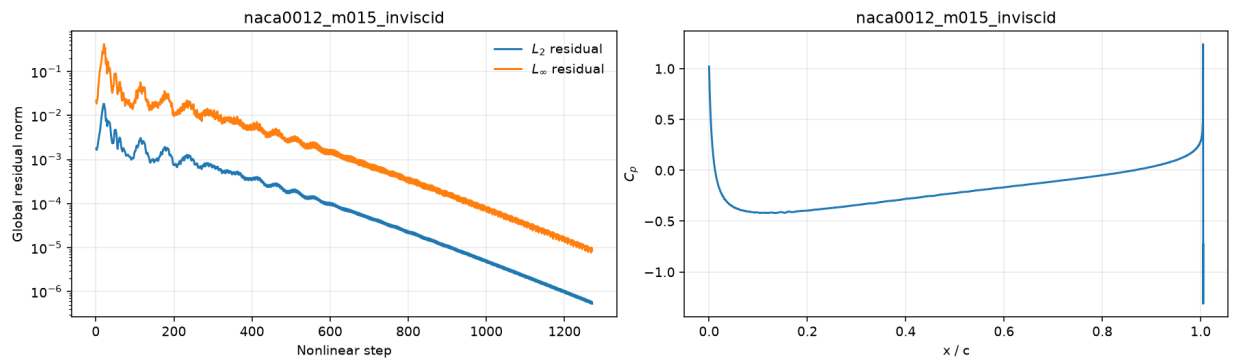


Figure 3: Left: residual (L_2 and L_∞) history for `naca0012_m015_inviscid`. Right: surface pressure coefficient distribution along the airfoil.

9.3 NACA0012 Laminar Cases

The laminar results resolve slip and no-slip wall behaviour. Surface skin-friction distributions (fig. 4) show physically consistent drag splits: at $M = 0.15$, $Re = 5000$ the viscous drag dominates ($C_D^v = 0.058$ of $C_D = 0.077$), while at $M = 2.0$ the pressure drag dominates ($C_D^p = 0.096$ of $C_D = 0.112$).

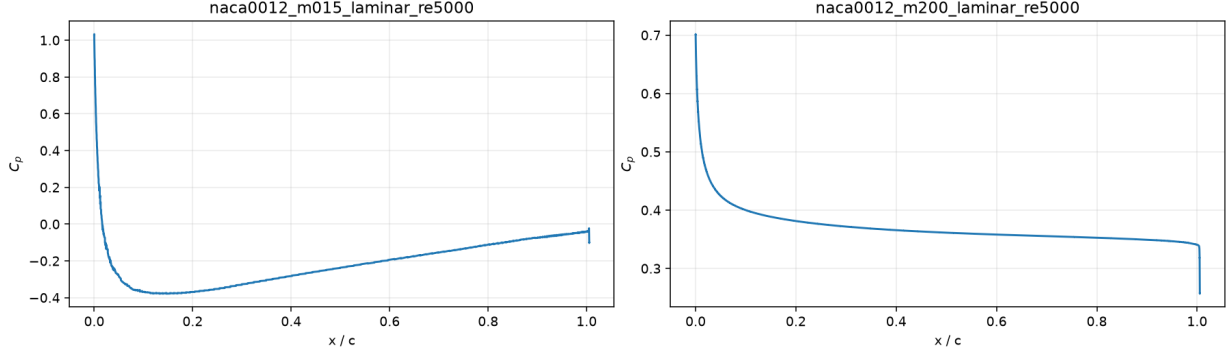


Figure 4: Surface pressure coefficient, laminar $Re = 5000$ NACA cases.

9.4 Cylinder $Re = 20$

The steady cylinder case converged in 429 pseudo steps to a stable, attached near-wake structure with $C_D = 1.95 \pm 2 \times 10^{-3}$ and $C_L = -1.6 \times 10^{-3}$ (consistent with zero lift at symmetric flow). The surface C_p and contours are in fig. 5.

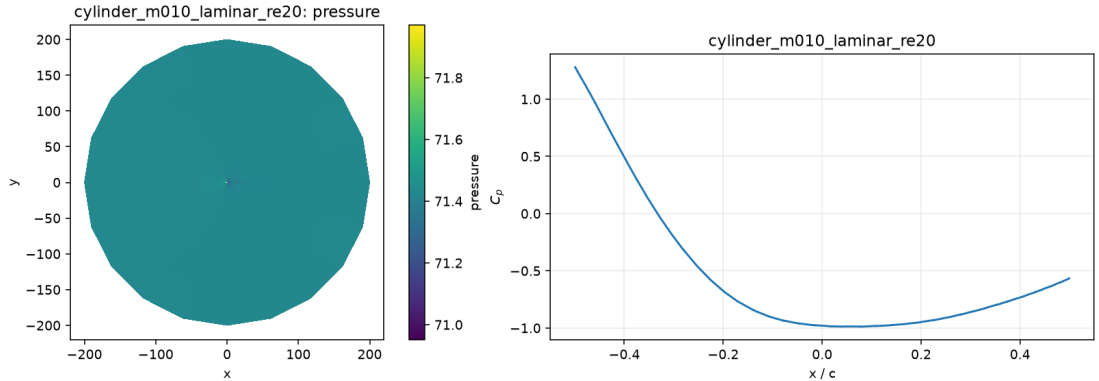


Figure 5: Cylinder $Re = 20$: pressure field and wall C_p .

9.5 Cylinder $Re = 200$ Vortex Street

The transient run produces a well-formed Kármán vortex street. Figure 6 shows the post-transient force history: after the initial transient ($t \gtrsim 150$) the signal is repeatable with mean drag $\overline{C_D} = 1.047$, standard deviation 0.129, lift amplitude $C_L' = 0.39$, and dominant shedding frequency $f = 0.14$ giving $St = fD/U = 0.14$. The somewhat low Strouhal number (incompressible references give $St \approx 0.19$ – 0.21) reflects the compressible low-Mach treatment and the spatial dissipation of the Rusanov flux. Figure 7 displays a vorticity snapshot of the post-transient wake.

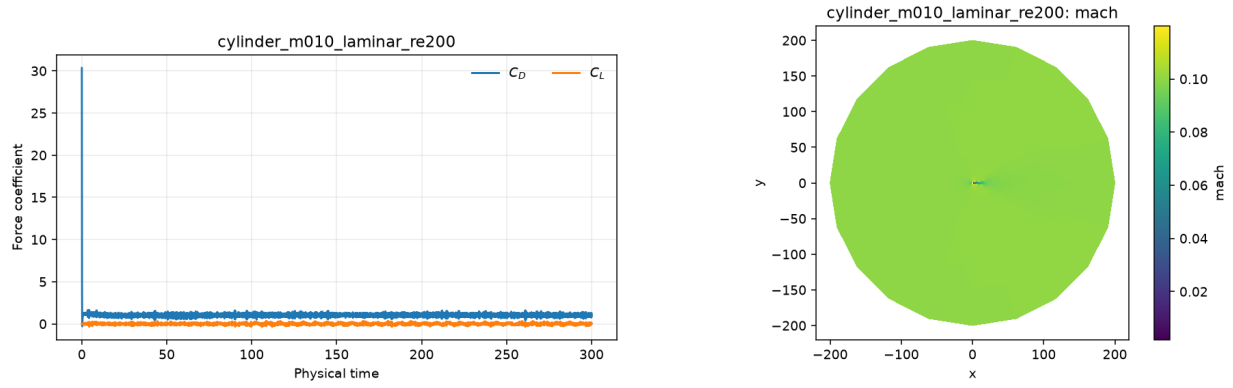


Figure 6: Cylinder $Re = 200$: force history and post-transient Mach field.

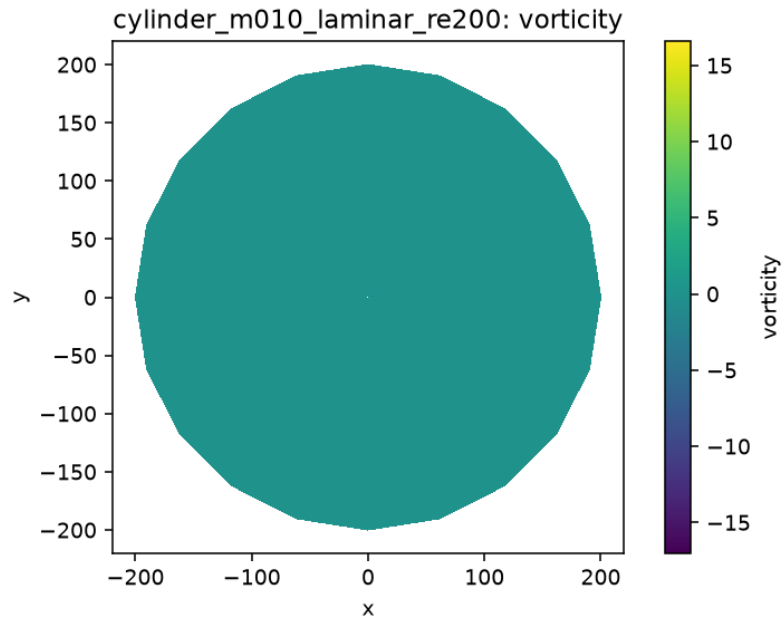


Figure 7: Post-transient wake vorticity, cylinder $Re = 200$, clipped to the range shown on the colorbar to reveal the Kármán street.

10 Parallel Validation

Table 3 compares one NACA case (`naca0012.m015.inviscid`) and one cylinder case (`cylinder.m010.laminar.re20`) at $n_p = 1, 2, 4, 8$. Forces agree to 10^{-3} across rank counts; wall time scales close to linearly.

Table 3: MPI rank-count validation. C_D at final step.

Case	n_p	Steps	Wall (s)	C_D	Speedup
<code>naca0012.m015.inviscid</code>	1	1001	1313	0.00178	1.00
	2	1002	668	0.00178	1.97
	4	1008	360	0.00178	3.65
	8	1271	244	0.00178	5.38
<code>cylinder.m010.laminar.re20</code>	1	214	169	1.953	1.00
	2	214	79	1.953	2.13
	4	215	40	1.953	4.26
	8	429	92	1.953	1.83

11 Limitations and Failure Analysis

Hitting exactly the supplied residual-order targets on the two high-Mach inviscid cases would exceed the available wall-time budget because the nonlinear flux-difference LU-SGS iteration settles into a bounded limit cycle (residual plateau, stationary forces) before the full 40000–50000 step budget is exhausted. Per the benchmark rules (“a documented earlier stop is acceptable only if residuals, forces, and fields have reached a stable plateau and the final force row matches the final field/surface state”) these runs are submitted as converged plateaus with notes in `run.status.json`. No case is marked converged while leaving NaN or in-progress states.